

☒ SESSION 15 – DETAILED THEORY

☒ 1. Libraries and Functional Programming in Python

☒ What are Libraries?

A **library** is a collection of pre-written code that can be reused. Python offers both **standard libraries** (built-in) and **third-party libraries** (via pip).

- **Standard Libraries:** math, os, random, datetime, etc.
- **Third-party Libraries:** requests, numpy, pandas, flask, etc.

☒ Importing Libraries

python

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```
import math print(math.sqrt(25)) # Output: 5.0
```

☒ Functional Programming

Functional programming is a paradigm where:

- **Functions are first-class objects** (can be passed around)
- **Immutability** is preferred
- Focuses on **pure functions** and **no side effects**

Core Tools:

- **lambda**: Anonymous functions
-

map(): Applies function to all items

-

filter(): Filters based on condition

-

reduce(): Applies rolling computation

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```
from functools import reduce  
nums = [1, 2, 3, 4]  
squares = list(map(lambda x: x**2, nums))  
evens = list(filter(lambda x: x % 2 == 0, nums))  
product = reduce(lambda x, y: x * y, nums)
```

❏ 2. Servers and Clients Architecture

❏ What is Client-Server Architecture?

A **client** sends a request; the **server** processes and responds.

- **Client:** A user-side application (e.g., browser, mobile app)
- **Server:** Hosts services and data (e.g., web, database)

☒ Types of Servers

- **Web Server:** Serves websites (e.g., Apache, Nginx)
- **Database Server:** Manages databases (e.g., MySQL)
- **FTP Server:** File transfer
- **Mail Server:** Sends/receives emails

☒ Communication

- Typically happens over TCP/IP
- Protocols like HTTP, FTP, SMTP govern the communication

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Client (Browser) ☒ Request ☒ Server (Web Server) ☒ Response ☒ Client

☒ Python Example (TCP)

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```
# Server import socket s = socket.socket() s.bind(('localhost', 9999)) s.listen(1) conn, addr = s.accept() conn.send(b"Hello from server") # Client
c = socket.socket() c.connect(('localhost', 9999)) print(c.recv(1024).decode())
```

☒ 3. Web Servers and Client Scripting

☒ Web Server

A **web server** handles HTTP requests and serves HTML or other content.

Popular examples:

- Apache
- Nginx
- Python's built-in http.server

bash

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```
# Start a simple web server in Python python3 -m http.server 8000
```

☒ ☒ Client-Side Scripting

- Code that runs **in the browser**
- Language: **JavaScript**
- Enhances interactivity, form validations, AJAX, animations

☒ Web App Flow:

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Browser (Client) ↔ HTML/CSS/JS ↔ Web Server ↔ Backend Python Code ↔ Database

Flask Example (Python Web Server)

python

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```
from flask import Flask app = Flask(__name__) @app.route('/') def home(): return "Welcome to my web server" app.run()
```